

Final Session Season 26



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Play!



Why Should Backend Developers Care?

AI is not just about building chatbots.

As backend developers, we can use AI to:

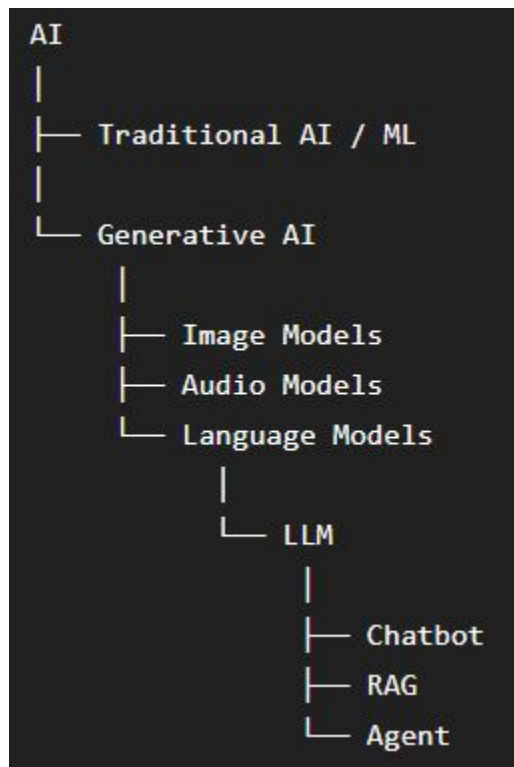
- Help in Writing and understand code faster
- Debug errors and investigate logs
- Generate and improve tests
- Understand documentation and codebases
- Search and analyze large amounts of information
- Build AI-powered backend features
- Automate repetitive engineering tasks

The goal

**Use AI to make us better
backend developers , not
replace our
understanding.**



AI vs ML vs GenAI vs LLM vs Chatbot vs Agent



AI vs ML vs GenAI vs LLM vs Chatbot vs Agent

Term	Simple meaning
AI	The broad field of making machines perform intelligent tasks
ML Model	A model that learned patterns from data to make predictions/decisions
Generative AI	AI that generates new content
Language Model	Model specialized in understanding/generating language
LLM	A large language model, e.g. GPT
Chatbot	An application/interface that uses a model to converse
RAG	Gives the model relevant external information before it answers
Agent	An AI system that can decide what actions/tools to use to accomplish a task

ChatGPT is an application. GPT is a model. An agent is a system built around a model.

Using AI as a Backend Developer

Coding

Before coding

- Understand requirements
- Break a feature into tasks
- Design APIs
- Suggest architecture

During coding

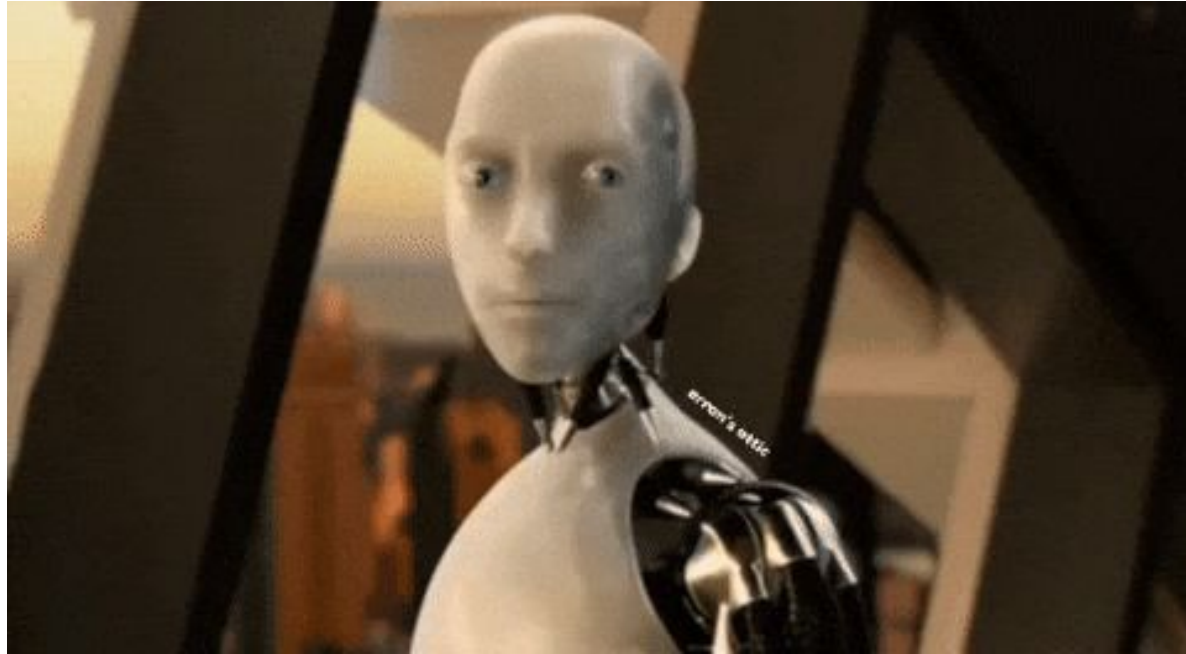
- Explain unfamiliar code
- Generate boilerplate
- Suggest implementations
- Refactor code

After coding

- Review your code
- Find edge cases
- Generate tests
- Improve documentation



We did not say "Write Code"



Building AI Features into Backend Systems

The AI model is just another service our backend communicates with.



Building AI Features into Backend Systems

AI Is Not Always an LLM

Different problems can need different AI approaches.

Problem	Possible approach
Generate text	LLM
Summarize document	LLM
Semantic search	Embeddings + Vector DB
Detect spam	ML / classifier
Recommend products	Recommendation model
Generate image	Image generation model
Speech → text	Speech model
Text → speech	Speech model

Agents



1. Question
2. LLM
3. Answer



1. Goal
2. Agent
3. Think about next step
4. Use tool
5. Observe result
6. Decide next step
7. Use another tool
8. Final result

What Is a Tool?

A tool is an action the AI is allowed to perform.

Tools:

- ★ `searchCode()`
- ★ `readFile()`
- ★ `getLogs()`
- ★ `queryDatabase()`
- ★ `searchDocs()`
- ★ `runTests()`

The agent doesn't magically have these abilities.

We explicitly give it access.

Agent examlpe

Option: Gemini Code Assist

[Gemini Code Assist for individuals](#)

It has a free individual tier and integrates directly into VS Code.

Or Install the extension

In VS Code: Search: Gemini Code Assist

then,

Open the **whole project**, not just one file.

Open Gemini Code Assist's **Agent Mode** in VS Code.

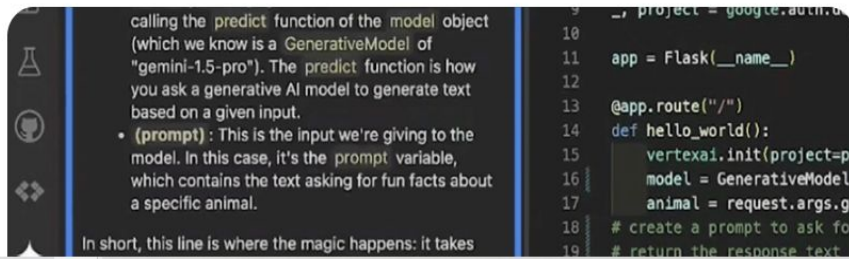


Gemini Code Assist: AI-first coding in your natural language

Uncap your potential and get all of your development done using Gemini Code Assist for individuals at no cost, or Gemini Code Assist for your business.

Try it now

you can get more done
with each **prompt**.



Experience coding with fewer limits in your favorite IDEs

Gemini Code Assist brings the power of Gemini 2.5 to your IDE at no cost (with Gemini 3 coming soon), with a chat interface that is aware of your code, automatic code completions as you write, and generation and transformation

Available in your terminal, favorite IDEs and platforms



Terminal, extensions and marketplaces



Google Cloud

Visual Studio Marketplace

Download Visual Studio Code extension



JetBrains Marketplace

Download the plugin



Terminal

Install Gemini CLI



GitHub Marketplace

Install the code review agent



Firebase

Use Gemini in Firebase for coding assistance



Android Studio

Use Gemini in Android Studio for coding assistance





Gemini Code Assist

[Google](#)  google.com |  5,357,520 installs |      (894) | Free

AI-assisted development powered by Gemini

Install

[Trouble Installing?](#) 

[Overview](#)

[Version History](#)

[Rating & Review](#)

Gemini Code Assist - Your AI assistant powered by Gemini

Bring the power of Gemini to help you build applications faster and easier than ever before.

Key Features

- Coding assistance powered by Gemini.
- Contextualized responses to your prompts to help guide you on what you're trying to do with your code.
- Source citations regarding which documentation and code samples the assistant used to generate its responses.
- Automated code reviews in GitHub with Gemini Code Assist for GitHub.
- Also available in Firebase, Android Studio, IntelliJ, Google Cloud Databases, BigQuery, Apigee, and more.



Gemini Code Assist

Google  google.com |  5,356,631 |     (894)

AI-assisted development powered by Gemini

Installing



Auto Update



DETAILS

FEATURES

Gemini Code Assist - Your AI assistant powered by Gemini

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Marketplace

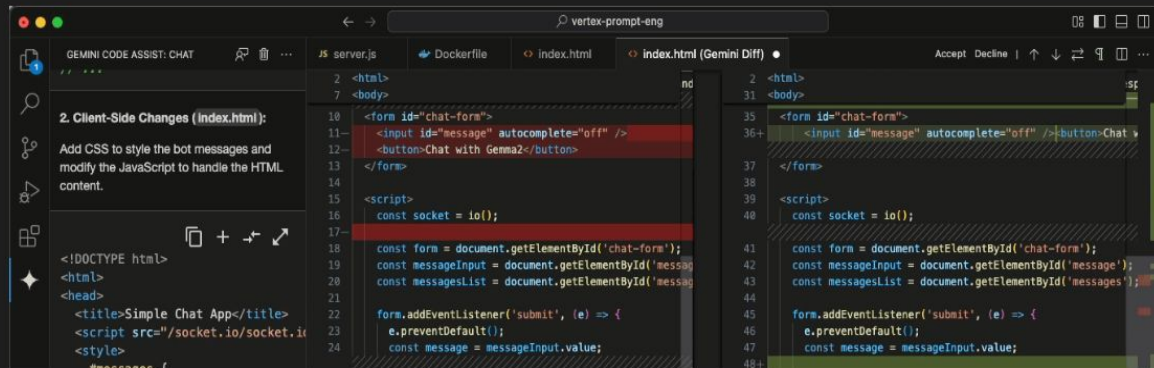
Identifier	google.geminicodeassist
Version	2.100.0
Published	2 years ago
Last Released	2 weeks ago

Categories

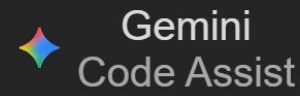
[Programming Languages](#) [Machine Learning](#) [AI](#) [Chat](#)

Resources

-  [License](#)
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Workspace is not trusted.
Some features may not work
as expected.



Supercharge your workflow with
AI-powered chat, code
completion, code generation, and
more.

Sign in

Get started with Gemini Code Assist

Boost your coding productivity and quality with intelligent code suggestions and chats powered by Gemini, supporting all programming languages in the public domain.

Write better code, faster

Get started with Gemini Code Assist's many features:

- Press **Control+I** to **transform your code** as well as execute your own **custom commands**
- Select comments as prompts in the editor and press **Control+Enter** to **generate code**
- Click the Gemini Logo (🔮) in the Activity Bar to use Gemini Code Assist **chat** to ask questions about and improve your code
- Add **files and folders** to your chat context
- Add custom **rules** to enhance chat responses

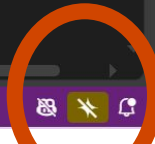
Review the latest Code Assist feature [releases](#)

```
func parseCatalog() []*pb.Product {  
    if reloadCatalog() { len(cat.Products) == 0 {  
        err := readCatalogFile(&cat)  
        if err != nil {  
            return []*pb.Product{}  
        }  
    }  
    return cat.Products  
}
```



Set up your organization for success

With Gemini Code Assist Standard and Enterprise editions, you can set up **logging and auditing**, see team productivity metrics, manage licensing and benefit from IP indemnification.



RAG

Suppose we ask:

"What is the authentication policy of our backend?"

The LLM may not know.

Why?

Because our private documentation isn't necessarily part of its available knowledge/context.

RAG in Backend

- User
- POST /ask
- Backend
- Search knowledge base
- Relevant documents
- LLM
- Answer

RAG vs Agent



Find information for
me.



Do something for me.

They can also work together

Security, Limitations & Human Control

AI Can Be Wrong, AI can:

- Hallucinate information
- Generate incorrect code
- Miss bugs
- Misunderstand requirements
- Produce insecure solutions
- Make confident mistakes

Therefore:

Never blindly trust AI-generated code or decisions.

Give Agents Minimum Permissions



- Agent
- Production Database
- READ + WRITE + DELETE



- Agent
- Read-only database
- Limited tools
- Human approval
- actions

Give AI only the permissions it actually needs.

Human in the Loop



AI Generate documentation
Automatic



AI Delete production data

The slide features four decorative corner elements. The top-left corner has a teal wavy shape, a yellow star, and a brown oval. The top-right corner has a blue U-shape with a red circle inside, surrounded by yellow dashes, and a purple teardrop. The bottom-left corner has a yellow flower, a brown teardrop, and a purple teardrop. The bottom-right corner has a purple flower, a brown teardrop, a yellow teardrop, and a blue star.

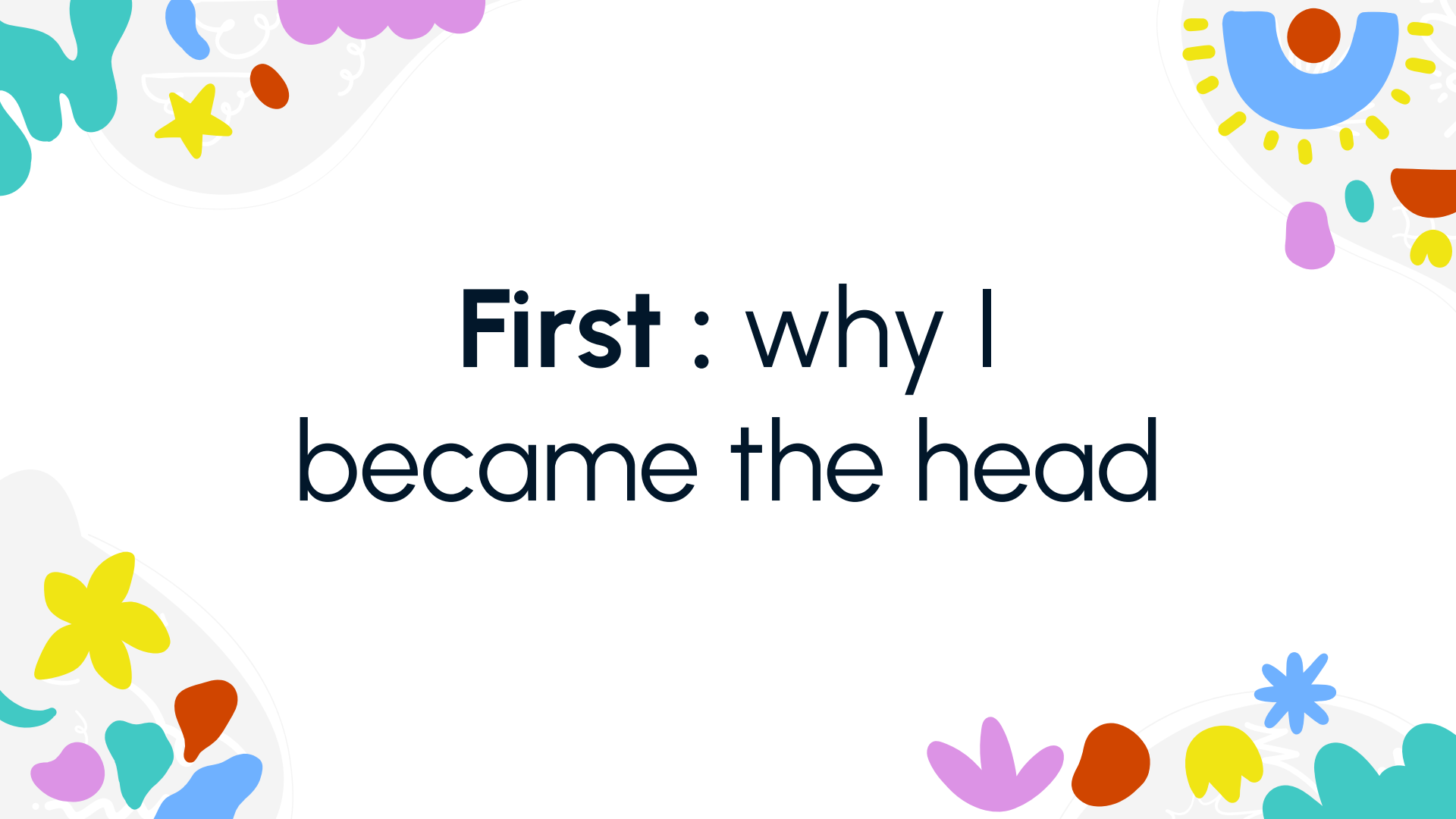
Thanks!

بهزر لسه مخلصناش طبعاً



The slide features four decorative corner elements. Top-left: A teal cloud-like shape, a blue swirl, a purple cloud-like shape, a yellow star, and a brown oval. Top-right: A blue U-shape with a red circle inside, surrounded by yellow dashes, with a purple oval, a teal oval, and a brown oval below. Bottom-left: A yellow flower, a brown oval, a purple oval, a teal shape, and a blue shape. Bottom-right: A purple flower, a brown oval, a yellow flower, a blue star, and a teal cloud-like shape.

**Let's remember our
season**

The slide features four decorative corner elements. Top-left: A teal cloud-like shape, a blue swirl, a purple cloud, a yellow star, and an orange dot. Top-right: A blue U-shape with a red circle inside, surrounded by yellow dashes, with a purple, teal, and orange shape below. Bottom-left: A yellow flower, a red teardrop, a purple shape, and a teal shape. Bottom-right: A purple flower, an orange teardrop, a yellow flower, and a blue star.

First : why I
became the head

The slide features a light gray background with decorative elements in the corners. The top-left corner has a teal abstract shape, a yellow star, and a purple cloud-like shape. The top-right corner has a blue U-shaped object with yellow dots and a red circle. The bottom-left corner has a yellow flower, a red teardrop, and a purple shape. The bottom-right corner has a blue star, a yellow shape, a red teardrop, and a teal abstract shape.

Second: My Great Board and how we worked

The slide features a light gray background with decorative elements in the corners. The top-left corner has a teal abstract shape, a yellow star, and a purple cloud-like shape. The top-right corner has a blue U-shape with yellow dots, a red semi-circle, and a purple teardrop. The bottom-left corner has a yellow flower, a red teardrop, and a purple teardrop. The bottom-right corner has a blue star, a red teardrop, a yellow teardrop, and a teal abstract shape.

Third: A Quick Review of the entire season

Top 1 Full-Stack Project



A little bit of Emotions



Wipe Your Tears



The slide features four decorative corner elements. Top-left: A teal cloud-like shape, a blue swirl, a purple cloud, a yellow star, and an orange oval. Top-right: A blue U-shape with a red circle inside, surrounded by yellow dashes, with a purple oval, teal oval, and orange semi-circle below. Bottom-left: A yellow flower, a red teardrop, a purple oval, and a teal shape. Bottom-right: A purple flower, an orange oval, a yellow flower, a blue star, and a teal cloud-like shape.

Anyways, let's
Introduce our lovely
new BOARD!



The slide features four decorative corner elements. The top-left corner has a teal wavy shape, a yellow star, and a brown oval. The top-right corner has a blue U-shape with a red circle inside, surrounded by yellow dashes, and a purple teardrop. The bottom-left corner has a yellow flower, a brown oval, and a purple teardrop. The bottom-right corner has a blue star, a purple teardrop, a brown oval, and a yellow teardrop.

The last Kahoot!

The slide features four decorative corner elements. The top-left corner has a teal cloud-like shape, a yellow star, and a purple cloud. The top-right corner has a blue U-shaped object with yellow dots, a red semi-circle, and a purple teardrop. The bottom-left corner has a yellow flower, a red teardrop, and a purple teardrop. The bottom-right corner has a blue star, a red teardrop, a yellow teardrop, and a teal cloud-like shape.

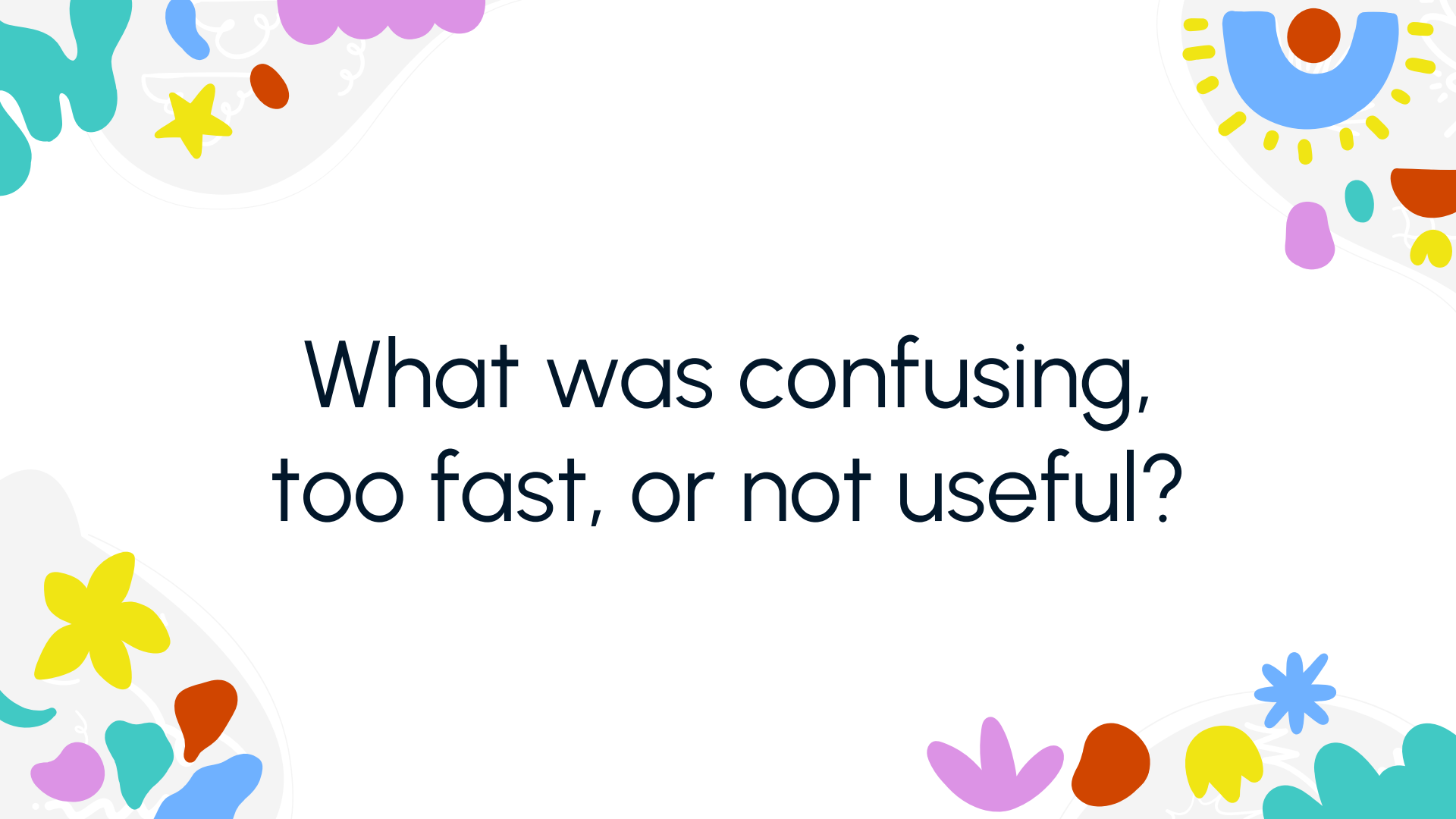
The session is over but
we will have a small
discussion

The image features a white background with four decorative corner elements. The top-left corner has a teal wavy shape, a yellow star, and a purple cloud-like shape. The top-right corner has a blue U-shaped object with yellow dots, a red semi-circle, and a purple teardrop. The bottom-left corner has a yellow flower, a red teardrop, and a purple teardrop. The bottom-right corner has a blue star, a red teardrop, a yellow teardrop, and a teal wavy shape.

What did you enjoy the
most this season?

The image features a central text area surrounded by a decorative border. The border consists of various colorful, abstract shapes and patterns, including a yellow star, a blue U-shape with yellow dots, a yellow flower, a blue star, and several colorful blobs in teal, purple, orange, and yellow. The background is white.

What was the most
useful session/topic for
you?

The slide features four decorative corner elements. Top-left: A teal cloud-like shape, a blue swirl, a purple cloud, a yellow star, and a brown oval. Top-right: A blue U-shape with a red circle inside, surrounded by yellow dashes, with a purple oval, teal oval, and red semi-circle below. Bottom-left: A yellow flower, a brown oval, a purple oval, a teal shape, and a blue shape. Bottom-right: A purple flower, a brown oval, a yellow flower, a blue star, and a teal cloud-like shape.

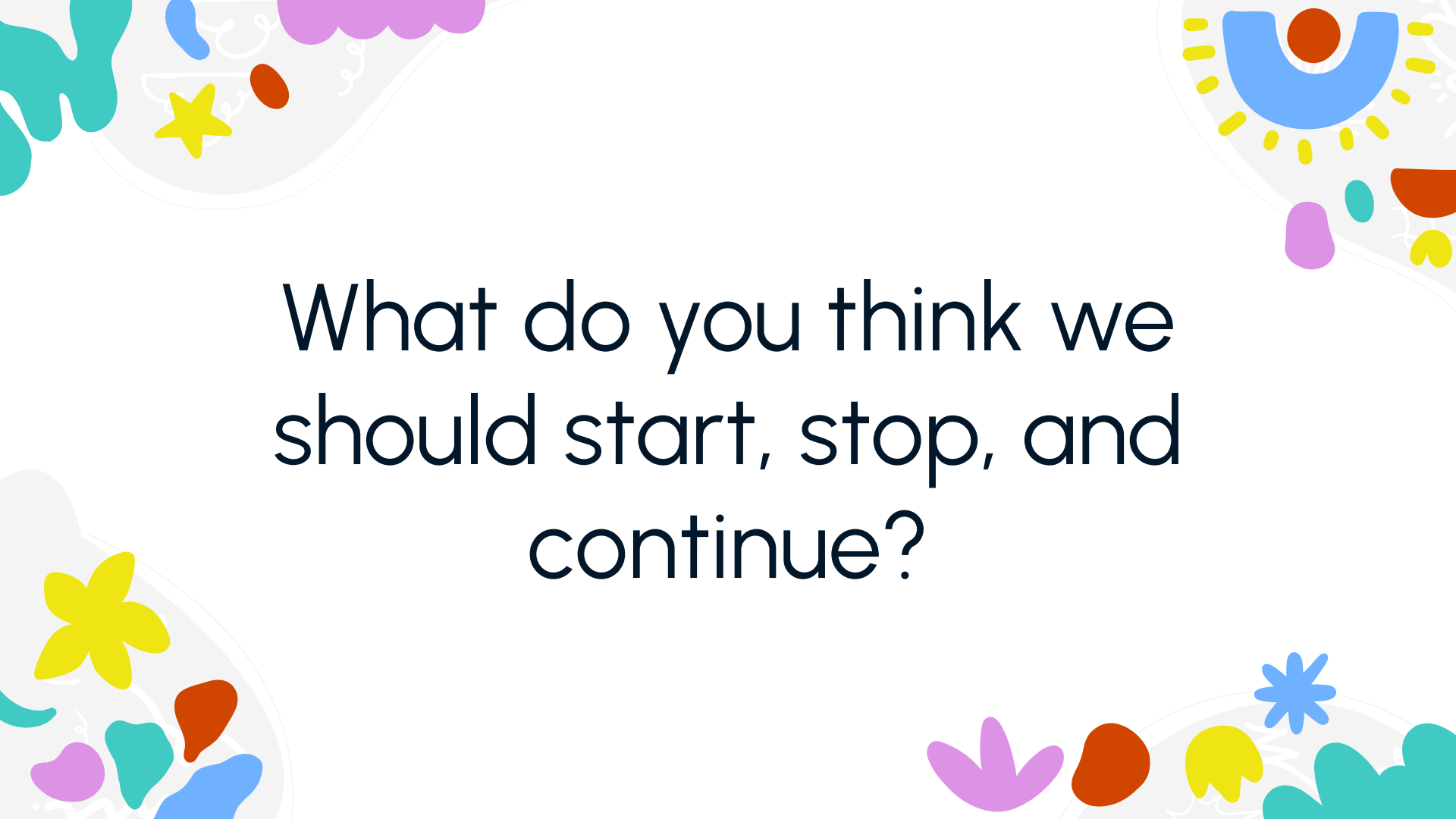
What was confusing,
too fast, or not useful?

The image features a white background with the text "What would you change if we ran the season again?" centered in a dark blue, sans-serif font. The corners are decorated with colorful, stylized floral and abstract shapes. The top-left corner has a teal shape, a yellow star, and a purple scalloped shape. The top-right corner has a blue U-shape with yellow dots, a red semi-circle, and a purple shape. The bottom-left corner has a yellow flower, a red shape, and a purple shape. The bottom-right corner has a blue star, a red shape, a yellow shape, and a teal shape.

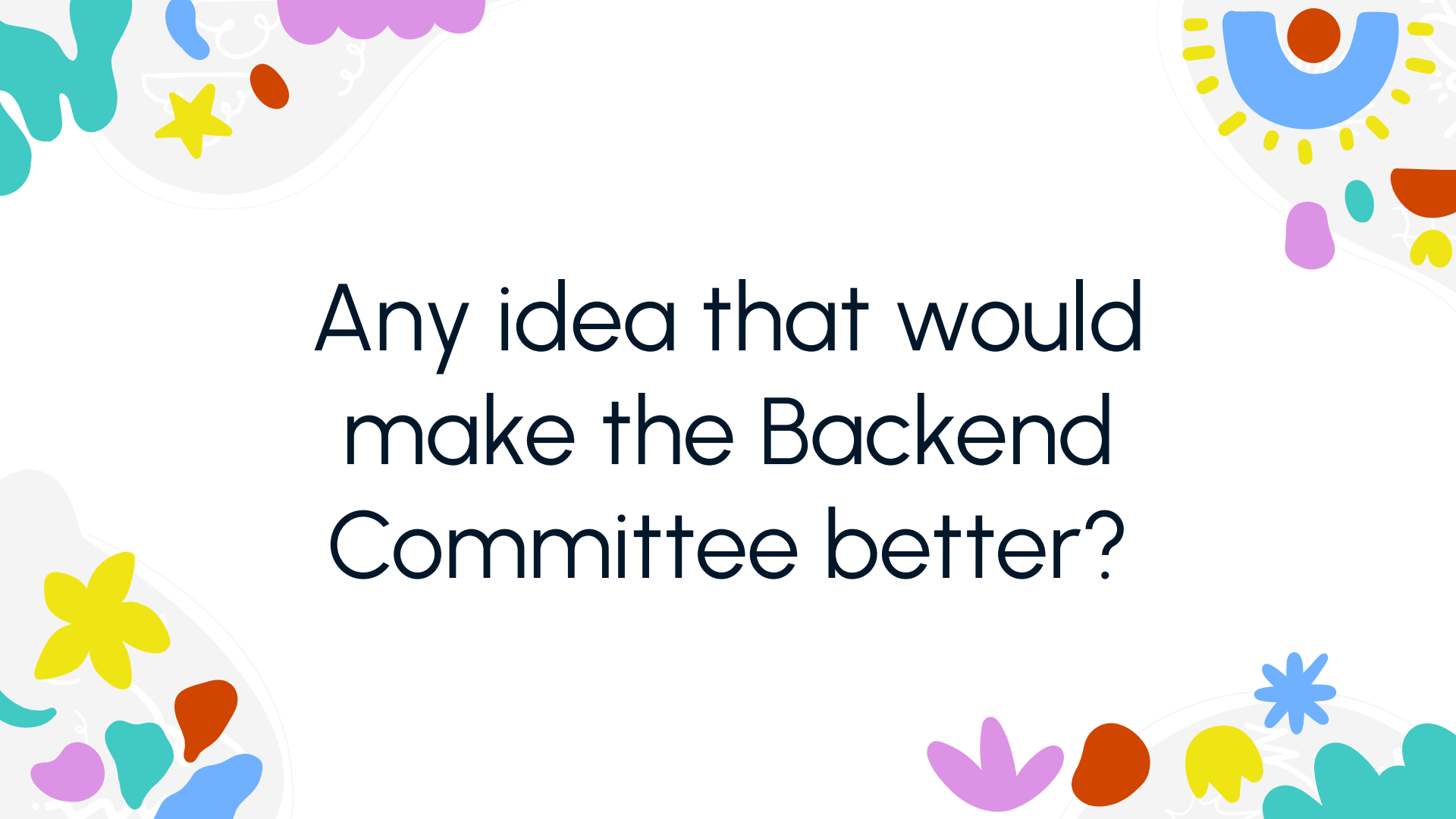
What would you
change if we ran the
season again?

The slide features four decorative corner graphics. The top-left corner has a teal cloud-like shape, a yellow star, and a brown oval. The top-right corner has a blue U-shape with yellow dots, a brown oval, a teal oval, and a yellow oval. The bottom-left corner has a yellow flower, a brown oval, a teal shape, and a purple shape. The bottom-right corner has a purple flower, a brown oval, a yellow flower, and a blue star.

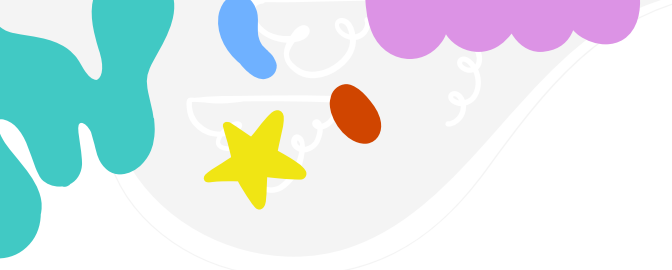
What topics do you
want to see next
season?

The slide features four decorative corner elements. Top-left: A teal cloud-like shape, a blue swirl, a purple cloud, a yellow star, and an orange oval. Top-right: A blue U-shape with a red circle inside, surrounded by yellow dashes, with a purple oval, teal oval, and orange semi-circle below. Bottom-left: A yellow flower, a red teardrop, a purple oval, a teal shape, and a blue shape. Bottom-right: A purple flower, an orange oval, a yellow flower, a blue star, and a teal cloud-like shape.

What do you think we
should start, stop, and
continue?



Any idea that would
make the Backend
Committee better?

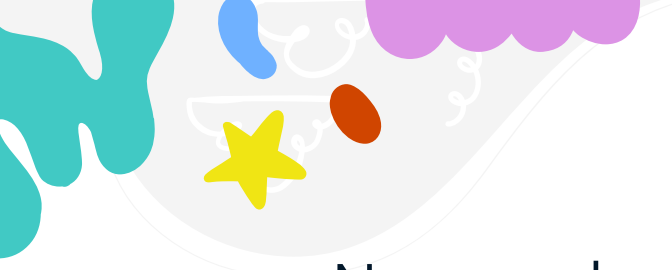


If you could change ONE thing
about the Backend Committee next
season, what would it be?

Be honest, this is for the next board
to learn from.



Any advice for the next
board? ❤️



Now each of you will have a piece of paper

your task is to make everyone in the room
type you a message or at least type their
name



And please type your name on top of the
paper.





Thanks!

بجد خلاص مش بهزر

